

YAO DAVID KOSSI

Quantitative Developer | Software Engineer

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LinkedIn | GitHub | Portfolio

PROFILE

Final-year Financial Engineering student at Grenoble INP – Ensimag, pursuing a dual degree in Quantitative Finance. Foundation in algorithms, stochastic modelling and software engineering, with hands-on experience in C++, C#, Python and production-oriented development. Seeking a **4-5 month** Quantitative Development / Software Engineering **internship** from **April or May 2027**. Mobile in Europe and Asia.

EDUCATION

Grenoble INP – Ensimag & Grenoble INP – IAE **Grenoble, France**
Engineering degree in Financial Engineering & Master's in Quantitative Finance *2026–2027*

Grenoble INP – Ensimag **Grenoble, France**
Engineering curriculum – Applied Mathematics, Computer Science and Financial Engineering *2024–2027*

Relevant coursework: algorithms and data structures, object-oriented programming, probability, stochastic processes, numerical methods, statistics, machine learning and quantitative finance.

TECHNICAL EXPERIENCE

Full-Stack Software Engineer Intern – Artelia, ARCADE project **Jun–Aug 2026**
Angular, TypeScript, C#/.NET, ASP.NET Core, PostgreSQL, AWS *Echirolles, France*

- Developed an internal data-catalogue platform end to end, from PostgreSQL modelling and a 50-endpoint REST API to the Angular frontend.
- Built and maintained 493 unit tests and 49 integration tests; containerised services and contributed to CI/CD delivery through Azure Pipelines, Amazon ECR and ECS.

SELECTED PROJECTS

Systematic Trading Strategy *C#/.NET | Aug–Sep 2026*
Designed and implemented an academic systematic-strategy application in .NET, applying quantitative signals to portfolio and trading decisions.

Derivatives Hedging and Option Pricing *C++ | Sep 2026*
Implemented pricing for European options with Black–Scholes and Monte Carlo simulation, together with a derivatives-hedging framework.

Portfolio Risk Modelling with GARCH, EVT and Copulas *Python | 2026*
Built a modular framework for VaR and Expected Shortfall using GARCH filtering, POT–GPD tails, multivariate copulas and Monte Carlo simulation; backtested forecasts with Kupiec and Christoffersen tests.

Deca Compiler *Java, ANTLR4 | Jan 2026*
Developed lexical and syntax analysis, contextual checks, code generation and automated tests for a Java-like language.

JPEG Decoder *C | May 2025*
Implemented Huffman decoding, bitstream manipulation, inverse DCT and memory-conscious image reconstruction.

Detailed project write-ups and demonstrations: david-kbless.github.io/projects

SKILLS

Programming: C++, C#, Python, C, Java, SQL, TypeScript, R
Tools: Linux, Git, GCC, Bash, Docker/Podman, AWS, Azure Pipelines
Quantitative: Black–Scholes, Monte Carlo, stochastic processes, option pricing, hedging, VaR/ES, GARCH, EVT, copulas
Languages: French (native), English (B1)

ADDITIONAL EXPERIENCE

Mathematics and Computer Science Tutor, Université Grenoble Alpes (Sep–Dec 2024) | WWF France donor